

Season 1 intro

How LLMs work.

What large language models (LLMs) actually do, from the first token to reasoning

What this season answers

- What does a large language model actually do?
- Why is everything counted in tokens?
- How do models represent meaning and decide what matters?
- Why do models lose information and make things up?
- When is a reasoning model worth the cost?

12 episodes

E1	One token at a time	
E2	Tokens	
E3	Embeddings	
E4	Attention	
E5	The transformer	
E6	Context windows	
E7	Lost in the middle	Field note
E8	Sampling	
E9	Hallucination	
E10	Reasoning versus standard	Field note
E11	When is a reasoning model worth the cost?	Decision
E12	Season 1 recap	Recap

Two field notes, with open code

S1 E7

Does a fact's position in a long document change whether a model finds it?

S1 E10

When does reasoning improve accuracy enough to justify the extra tokens and time?

Measured across three model providers, with results you can rerun.

Code and results: github.com/BrendanKellyAI/applied-ai-lab

Next: S1 E1 One token at a time

What an LLM actually does, and why every call starts fresh.

Where this fits

The series map

S1 LLMs	S2 Retrieval	S3 Strategy
S4 Adapting	S5 Agents	S6 Economics
S7 Data	S8 Evaluation	S9 AI teams
S10 Reliability	S11 Security	S12 Delivery
S13 Performance	S14 Resilience	S15 Governance

☐ Leadership season

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 Episode guide	brendankellyai.github.io
 Code	github.com/BrendanKellyAI/applied-ai-lab